Equilibrium Simulation: Mathematical Framework

Magnani & Munro (2020) with Treatment-Specific Payoffs

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1 Model

We adapt the dynamic market runs model of Magnani & Munro (2020) to our experimental design. The game consists of $N = 4$ risk-averse investors, each holding one unit of an asset. Time is discrete, and the game terminates each period with probability λ. Investors observe public signals about a binary state $z \in \{G, B\}$ and decide whether to sell irreversibly. Our innovation is to compare two payoff treatments—*Random* and *Average*—that differ in how simultaneous sellers split the available prices.

1.1 Notation

Symbol	Definition
$N = 4$	Number of investors
$n \in \{1, 2, 3, 4\}$	Number of investors still holding
$\pi \in (0, 1)$	Posterior probability that state is Good, $\Pr(z = G)$
$\sigma(n, \pi)$	Symmetric mixing probability: each holder sells with prob σ
$V(n, \pi)$	Value for a holder at start of period (before signal)
$W(n, \pi')$	Value for a holder after signal observation (at decision point)
$\lambda = 0.125$	Per-period termination probability
$v = 20$	Asset return in good state
$p_n = 2n$	Price when n holders remain: $p_1 = 2,\ p_2 = 4,\ p_3 = 6,\ p_4 = 8$
$\mu_B = 0.675$	$\Pr(\text{bad signal} \mid z = B)$
$\mu_G = 0.325$	$\Pr(\text{bad signal} \mid z = G)$
α	CRRA risk aversion parameter
$u(x) = \frac{x^{1-\alpha}}{1-\alpha}$	CRRA utility ($u(x) = x$ if $\alpha = 0$; $u(x) = \ln x$ if $\alpha = 1$)

1.2 Information structure

Agents start with common prior $\pi_0 = 0.5$. Each period, a public signal $y \in \{g, b\}$ is drawn with accuracy $\mu_B + \mu_G = 1$. We adopt the Magnani & Munro normalization so that good and bad

signals are exact Bayesian inverses. The posterior updates are:

$$\pi_g(\pi) = \frac{\pi(1-\mu_G)}{\pi(1-\mu_G) + (1-\pi)\mu_G} \qquad \text{(after good signal)} \tag{1}$$

$$\pi_b(\pi) = \frac{\pi(1-\mu_B)}{\pi(1-\mu_B) + (1-\pi)\mu_B} \qquad \text{(after bad signal)} \tag{2}$$

The unconditional signal probabilities given belief π are:

$$\Pr(y = g \mid \pi) = (1-\mu_G)\pi + \mu_G(1-\pi) \tag{3}$$

$$\Pr(y = b \mid \pi) = (1-\mu_B)\pi + \mu_B(1-\pi) \tag{4}$$

1.3 Belief grid

Since $\mu_B + \mu_G = 1$, the likelihood ratio update from a bad signal is the exact multiplicative inverse of a good signal:

$$\frac{\Pr(y = b \mid B)}{\Pr(y = b \mid G)} = \frac{\mu_B}{\mu_G} = \frac{0.675}{0.325} \approx 2.077 = \left(\frac{\mu_G}{\mu_B}\right)^{-1} = \left(\frac{\Pr(y = g \mid B)}{\Pr(y = g \mid G)}\right)^{-1}$$

Consequently, after k bad and j good signals from $\pi_0 = 0.5$, the posterior depends only on the net bad count $d = k - j$:

$$\pi(d) = \frac{1}{1 + \left(\frac{\mu_B}{\mu_G}\right)^d}$$

The reachable belief grid consists of $2T_{\max}+1$ values, computed by applying $d \in \{-T_{\max}, \ldots, T_{\max}\}$. We set $T_{\max} = 20$, yielding a 41-point grid spanning from $\pi(-20) \approx 4.5 \times 10^{-7}$ to $\pi(20) \approx 1 - 4.5 \times 10^{-7}$.

2 Payoff structure

2.1 Voluntary selling

When k investors sell simultaneously from n remaining holders, the prices consumed are $\{p_n, p_{n-1}, \ldots, p_{n-k+1}\}$. The payoff depends on the treatment:

Random treatment. Each seller is randomly assigned one of the k positions with equal probability, receiving the corresponding price:

$$\rho_{\text{random}}(n,k) = \frac{1}{k}\sum_{j=0}^{k-1} u(p_{n-j}) \tag{5}$$

Average treatment. Each seller receives the average price:

$$\rho_{\text{average}}(n,k) = u\left(\frac{1}{k}\sum_{j=0}^{k-1} p_{n-j}\right) \tag{6}$$

For $k = 1$, both treatments are identical: $\rho(n,1) = u(p_n)$. For $k > 1$ and $\alpha > 0$, Jensen's inequality gives $\rho_{\text{average}} > \rho_{\text{random}}$ since u is strictly concave.

2.2 Forced liquidation

If the game terminates in the bad state with m holders remaining, all are forced to sell at prices $\{p_1, \ldots, p_m\}$:

Random: $L_{\text{random}}(m) = \dfrac{1}{m}\sum_{i=1}^{m} u(p_i)$ **Average:** $L_{\text{average}}(m) = u\left(\dfrac{1}{m}\sum_{i=1}^{m} p_i\right)$

Again, $L_{\text{average}}(m) \geq L_{\text{random}}(m)$ by Jensen's inequality for $\alpha > 0$.

3 Bellman equations

3.1 Pre-signal value

At the start of a period, before the signal is observed:

$$V(n, \pi) = \Pr(y{=}g \mid \pi) \cdot W\big(n, \pi_g(\pi)\big) + \Pr(y{=}b \mid \pi) \cdot W\big(n, \pi_b(\pi)\big) \tag{7}$$

3.2 Post-signal value

After observing the signal and updating to π', each holder sells with probability $\sigma = \sigma(n, \pi')$:

$$W(n, \pi') = \sigma \cdot U_{\text{sell}}(n, \sigma) + (1 - \sigma) \cdot U_{\text{hold}}(n, \pi', \sigma) \tag{8}$$

3.3 Expected utility of selling

The investor sells along with j of the other $n-1$ holders (each selling independently with probability σ):

$$U_{\text{sell}}(n, \sigma) = \sum_{j=0}^{n-1} \binom{n-1}{j} \sigma^j (1 - \sigma)^{n-1-j} \cdot \rho(n,\, j + 1) \tag{9}$$

Note that U_{sell} does not depend on π'—the selling payoff is realized immediately and does not depend on the state.

3.4 Expected utility of holding

$$U_{\text{hold}}(n, \pi', \sigma) = \sum_{j=0}^{n-1} \binom{n-1}{j} \sigma^j (1 - \sigma)^{n-1-j} \cdot H(n{-}j,\, \pi') \tag{10}$$

3.5 Continuation value

After j others sell, $m = n - j$ holders remain. With probability λ the game terminates (good state yields v, bad state forces liquidation); with probability $1 - \lambda$ the game continues:

$$H(m, \pi') = \lambda\big[\pi' \cdot u(v) + (1 - \pi') \cdot L(m)\big] + (1 - \lambda) \cdot V(m, \pi') \tag{11}$$

4 Equilibrium

We solve for symmetric Markov-perfect equilibria: $\sigma_i(n, \pi) = \sigma(n, \pi)$ for all investors i.

4.1 Equilibrium conditions

For each state (n, π'):

1. If $U_{\text{sell}}(n, 0) \leq U_{\text{hold}}(n, \pi', 0)$: $\sigma = 0$ (holding dominates)
2. If $U_{\text{sell}}(n, 1) \geq U_{\text{hold}}(n, \pi', 1)$: $\sigma = 1$ (selling dominates)
3. Otherwise: $\sigma \in (0, 1)$ solves $U_{\text{sell}}(n, \sigma) = U_{\text{hold}}(n, \pi', \sigma)$ (indifference)

4.2 Boundary condition

For $n = 1$ (last holder): $\sigma(1, \pi) = 0$ for all π. Selling yields $u(p_1) = u(2)$, while holding preserves the option value $H(1, \pi') = \lambda[\pi' u(v) + (1 - \pi')u(p_1)] + (1 - \lambda)V(1, \pi')$. Since $u(v) = u(20) \gg u(2)$, holding always dominates when there is no strategic interaction.

4.3 Uniqueness

Proposition 1. *For each state (n, π') with $n \geq 2$, the equilibrium mixing probability σ^* is unique.*

Argument. Define $\Delta(\sigma) \equiv U_{\text{sell}}(n, \sigma) - U_{\text{hold}}(n, \pi', \sigma)$. The equilibrium condition requires $\Delta(\sigma^*) = 0$ (or a corner solution). We argue Δ is strictly decreasing in σ:

U_{sell} is strictly decreasing in σ. The selling payoff is $\rho(n, j+1)$ where $j \sim \text{Binomial}(n-1, \sigma)$ is the number of others who sell simultaneously. Since $\rho(n, k)$ is decreasing in k (more simultaneous sellers means each is more likely to receive a lower price), increasing σ shifts the distribution of j upward in the sense of first-order stochastic dominance, reducing U_{sell}.

U_{hold} is also decreasing in σ, but more slowly. When σ increases, the holder transitions to a state with fewer remaining holders ($m = n - j$ lower in expectation). Recall the continuation value:

$$H(m, \pi') = \lambda[\pi' u(v) + (1 - \pi')L(m)] + (1 - \lambda)V(m, \pi')$$

The good-state terminal payoff $u(v)$ does not depend on m. The forced-liquidation value $L(m)$ is increasing in m since the price schedule $p_m = 2m$ is increasing, so fewer holders means worse liquidation outcomes. The continuation value $V(m, \pi')$ is also increasing in m: lower m means lower available selling prices ($p_m = 2m$) and lower forced-liquidation payoffs in all future periods. Thus $H(m, \pi')$ is increasing in m, and higher σ (which shifts weight toward lower m) reduces U_{hold}.

Both utilities decrease, but U_{sell} decreases faster. Since $\Delta(\sigma) = U_{\text{sell}} - U_{\text{hold}}$ and both terms decrease with σ, uniqueness requires that U_{sell} is more sensitive to simultaneous competition than U_{hold}. Intuitively, a seller is *directly* affected by the number of simultaneous sellers (splitting the current prices), while a holder is only *indirectly* affected through future continuation values. We verify numerically that $\Delta(\sigma)$ is strictly decreasing at every grid point (n, π') across all parameter configurations (α, treatment), confirming at most one root in $(0, 1)$.

This yields a unique equilibrium σ^* at each state—a *strategic substitutes* property that rules out the self-reinforcing panics arising in bank run models with strategic complements. □

4.4 Threshold structure

For each $n \geq 2$, define $\pi^*(n)$ as the belief where $\Delta(0) = U_{\text{sell}}(n, 0) - U_{\text{hold}}(n, \pi^*, 0) = 0$. Then:

- For $\pi > \pi^*(n)$: $\sigma(n, \pi) = 0$ (pure hold)
- For $\pi \leq \pi^*(n)$: $\sigma(n, \pi) > 0$ (mixing or pure sell)

- $\sigma(n, \pi)$ is weakly decreasing in π (more pessimistic beliefs $\Rightarrow$ more selling)
- $\pi^*(n)$ is increasing in n (more holders $\Rightarrow$ selling starts at higher beliefs)

5 Treatment effects

Under risk neutrality ($\alpha = 0$), $u(x) = x$ is linear, so Jensen's inequality has no bite: $\rho_{\text{random}} = \rho_{\text{average}}$ and $L_{\text{random}} = L_{\text{average}}$. The equilibrium is identical across treatments.

Under risk aversion ($\alpha > 0$), two competing channels arise:

1. **Selling channel**: $\rho_{\text{average}} > \rho_{\text{random}}$ makes selling more attractive in the Average treatment.
2. **Holding channel**: $L_{\text{average}} > L_{\text{random}}$ makes forced liquidation less painful, increasing the value of holding.

Quantitatively, the holding channel dominates: thresholds $\pi^*(n)$ are lower under Average (narrower selling region). However, conditional on the same belief, the mixing probability σ is higher under Average (the selling payoff improvement dominates locally). These two effects partially offset.

6 Solution algorithm

The equilibrium has a circular structure: the value function $V(n, \pi)$ depends on the equilibrium strategies $\sigma(n, \pi)$, but the strategies themselves depend on the value function through the continuation payoff U_{hold}. We resolve this via value iteration, which repeatedly alternates between updating strategies (given the current value function) and updating the value function (given the current strategies).

6.1 Initialization

Construct the 41-point belief grid (Section 1.3). Store $V(n, \pi)$ and $\sigma(n, \pi)$ as arrays of dimension 4×41 (one entry per (n, π) pair). Initialize $V^{(0)}(n, \pi) = 0$ for all (n, π).

6.2 Iteration

Each iteration t proceeds in two passes:

Pass 1: Update σ and compute W. For each grid point (n, π'):

1. Compute $U_{\text{sell}}(n, 0)$ using Eq. (9). This does not depend on V since selling payoffs are immediate.
2. Compute $U_{\text{hold}}(n, \pi', 0)$ using Eq. (10) with $\sigma = 0$. This *does* depend on $V^{(t)}$ through the continuation value $H(m, \pi')$ in Eq. (11). Since $\sigma = 0$, nobody else sells, so $m = n$ and $U_{\text{hold}} = H(n, \pi')$.
3. **Find $\sigma^{(t+1)}$.** Define $\Delta(\sigma) \equiv U_{\text{sell}}(n, \sigma) - U_{\text{hold}}(n, \pi', \sigma)$, where U_{hold} uses $V^{(t)}$ in the continuation value. Then:

$$\sigma^{(t+1)} = \begin{cases} 0 & \text{if } \Delta(0) \leq 0 \\ 1 & \text{if } \Delta(0) > 0 \text{ and } \Delta(1) \geq 0 \\ \sigma^* \text{ s.t. } \Delta(\sigma^*) = 0 & \text{if } \Delta(0) > 0 \text{ and } \Delta(1) < 0 \end{cases}$$

The first case: even when no one else sells, holding weakly dominates, so no one sells. The second case: even when everyone else sells, selling still dominates, so everyone sells. The third case: Δ changes sign on $(0, 1)$, so an interior root σ^* exists; find it via Brent's method.

4. Compute $W^{(t+1)}(n, \pi') = \sigma \cdot U_{\text{sell}}(n, \sigma) + (1 - \sigma) \cdot U_{\text{hold}}(n, \pi', \sigma)$ using the σ just found.

Pass 2: Update V. For each grid point (n, π):

1. Compute the updated beliefs $\pi_g(\pi)$ and $\pi_b(\pi)$ via Eqs. (1)–(2).
2. Look up $W^{(t+1)}(n, \pi_g)$ and $W^{(t+1)}(n, \pi_b)$ by linearly interpolating the W array at the updated beliefs.
3. Set $V^{(t+1)}(n, \pi) = \Pr(y{=}g \mid \pi) \cdot W(n, \pi_g) + \Pr(y{=}b \mid \pi) \cdot W(n, \pi_b)$ per Eq. (7).

6.3 Convergence

Repeat until $\max_{n,\pi} |V^{(t+1)}(n, \pi) - V^{(t)}(n, \pi)| < 10^{-8}$. The output is the converged strategy table $\sigma(n, \pi)$ for all 4×41 state pairs.

7 Simulation engine

The equilibrium solution gives a strategy table $\sigma(n, \pi)$—the probability of selling at each state. To produce predictions comparable to experimental data, we need to know when sales actually occur. This requires simulation because the *timing* of sales depends on the stochastic signal process: beliefs random-walk through the grid, and a sale happens only when (a) the belief enters the selling region *and* (b) the coin flip with probability σ comes up "sell." Neither the threshold π^* nor the mixing probability σ alone tells us the average belief at sale.

7.1 One simulated game

For a given $(\alpha, \text{treatment})$ pair with converged strategy table $\sigma(n, \pi)$:

1. **Setup.** Set $n = 4$ holders, belief $\pi = 0.5$. Draw the true state $z \in \{G, B\}$ with equal probability. Initialize seller count $s = 0$.
2. **Each period** (repeat until $n = 1$ or game terminates):
   (a) *Signal.* Draw a signal from the true state: if $z = B$, the signal is bad with probability $\mu_B = 0.675$; if $z = G$, with probability $\mu_G = 0.325$. Update π via Bayesian rule (Eqs. (1)–(2)).
   (b) *Selling decisions.* Look up $\sigma(n, \pi)$ from the strategy table (interpolating if π falls between grid points). Each of the n holders independently draws a Bernoulli(σ): sell if 1, hold if 0. Let k be the total number who sell.
   (c) *Record sales.* If $k > 0$: for each of the k sellers, record the current belief π as the belief at seller position $s + 1, s + 2, \ldots, s + k$. Update $s \leftarrow s + k$ and $n \leftarrow n - k$.
   (d) *Termination.* With probability $\lambda = 0.125$, the game ends. If the game terminates, stop—any remaining holders did not sell voluntarily.

Note that multiple investors can sell in the same period (simultaneous sales), and they all receive the same belief π for their seller position. Games where the market terminates before any voluntary sale contribute no observations.

7.2 Aggregation

We run 10,000 games per $(\alpha, \text{treatment})$ pair. For each seller position $k \in \{1, 2, 3\}$, we compute the average belief at the time of sale across all games where that sale occurred:

$$\bar{\pi}_k = \frac{1}{|\mathcal{G}_k|} \sum_{g \in \mathcal{G}_k} \pi_k^{(g)}$$

where $\mathcal{G}_k$ is the set of games in which a kth sale happened and $\pi_k^{(g)}$ is the belief at that sale. The 4th investor never sells in equilibrium ($\sigma(1, \pi) = 0$ for all π), so we report three seller positions.

Following Magnani & Munro (2020, Table 2), we convert to $\Pr(z = B) = 1 - \pi$ for reporting.

8 Results

Table 1 reports the equilibrium-predicted average $\Pr(z = B)$ at the time of each sale across treatments and risk aversion levels, from 10,000 simulated games per cell.

	Random			Average		
α	1st seller	2nd seller	3rd seller	1st seller	2nd seller	3rd seller
0	0.812	0.813	0.825	0.812	0.813	0.825
0.1	0.812	0.812	0.813	0.812	0.812	0.812
0.2	0.740	0.802	0.832	0.751	0.807	0.829
0.3	0.705	0.767	0.828	0.709	0.775	0.826
0.4	0.686	0.725	0.800	0.685	0.723	0.797
0.5	0.678	0.698	0.759	0.677	0.695	0.752
0.6	0.676	0.685	0.723	0.675	0.682	0.712
0.7	0.675	0.678	0.697	0.675	0.675	0.679
0.8	0.672	0.676	0.681	0.675	0.675	0.675
0.9	0.641	0.667	0.681	0.669	0.675	0.676

Note: Each cell reports the average π (probability of bad state) at the time of the kth sale, from 10,000 simulated games using equilibrium strategies. At $\alpha = 0$ (risk neutral), predictions are identical across treatments. The 4th investor never sells in equilibrium ($n = 1$ boundary).

Table 1: Equilibrium predictions: average $\Pr(z = B)$ at each seller position. The 4th investor never sells ($n = 1$ boundary condition).

8.1 Validation against Magnani & Munro (2020)

Table 2 of Magnani & Munro reports the average $\Pr(z = B)$ at the first sale for their HIGH information environment ($\mu_B = 0.675$, NoCB). Our first-seller column reproduces their values:

	Our model	M&M Table 2	Difference
$\alpha = 0$ (risk neutral)	0.812	0.800	+0.012
$\alpha = 0.5$ (risk averse)	0.678	0.678	< 0.001

The risk-averse prediction matches exactly. The 1.2pp difference under risk neutrality is attributable to grid resolution: our 41-point grid produces slightly different value function interpolation than Magnani & Munro's numerical solution.